

Abstract

Nb-based MXenes, such as $\text{Nb}_4\text{C}_3\text{T}_x$, exhibit high electrical conductivity, making them well-suited for energy storage and electronic applications. However, the greater stability of Nb-based MAX phases poses challenges in the etching and, consequently, is difficult to prepare as MXenes. Additionally, Nb-based MXenes are highly susceptible to oxidation, altering their fundamental properties. This study focuses on $\text{Nb}_4\text{C}_3\text{T}_x$ MXenes, investigating their structural and thermal properties using XRD, SEM, Raman spectroscopy, TGA, and DSC measurements. Thermal analysis plays a crucial role in determining materials' structural integrity and potential applications under high-temperature conditions. XRD analysis confirmed the successful etching of aluminum, as evidenced by the disappearance of the characteristic aluminum peak, validating the preparation process. Raman spectra show significant peaks at ~ 300 and 677 cm^{-1} belonging to Nb-O and Nb-C respectively. TGA was utilized to measure thermal stability, oxidation analysis, and degradation behavior by monitoring the critical change in weight percentage. DSC measurements indicate no significant thermal events up to 300°C during heating and cooling cycles. Overall, this analysis aims to contribute to the information database of properties to extend the possible applications of $\text{Nb}_4\text{C}_3\text{T}_x$. This comprehensive analysis provides valuable insights into the structural and thermal behavior of $\text{Nb}_4\text{C}_3\text{T}_x$ MXenes, contributing to the knowledge base necessary for their potential application in high-temperature and oxidation-sensitive environments. The findings aim to expand the scope of Nb-based MXenes for future technological advancements.

Introduction

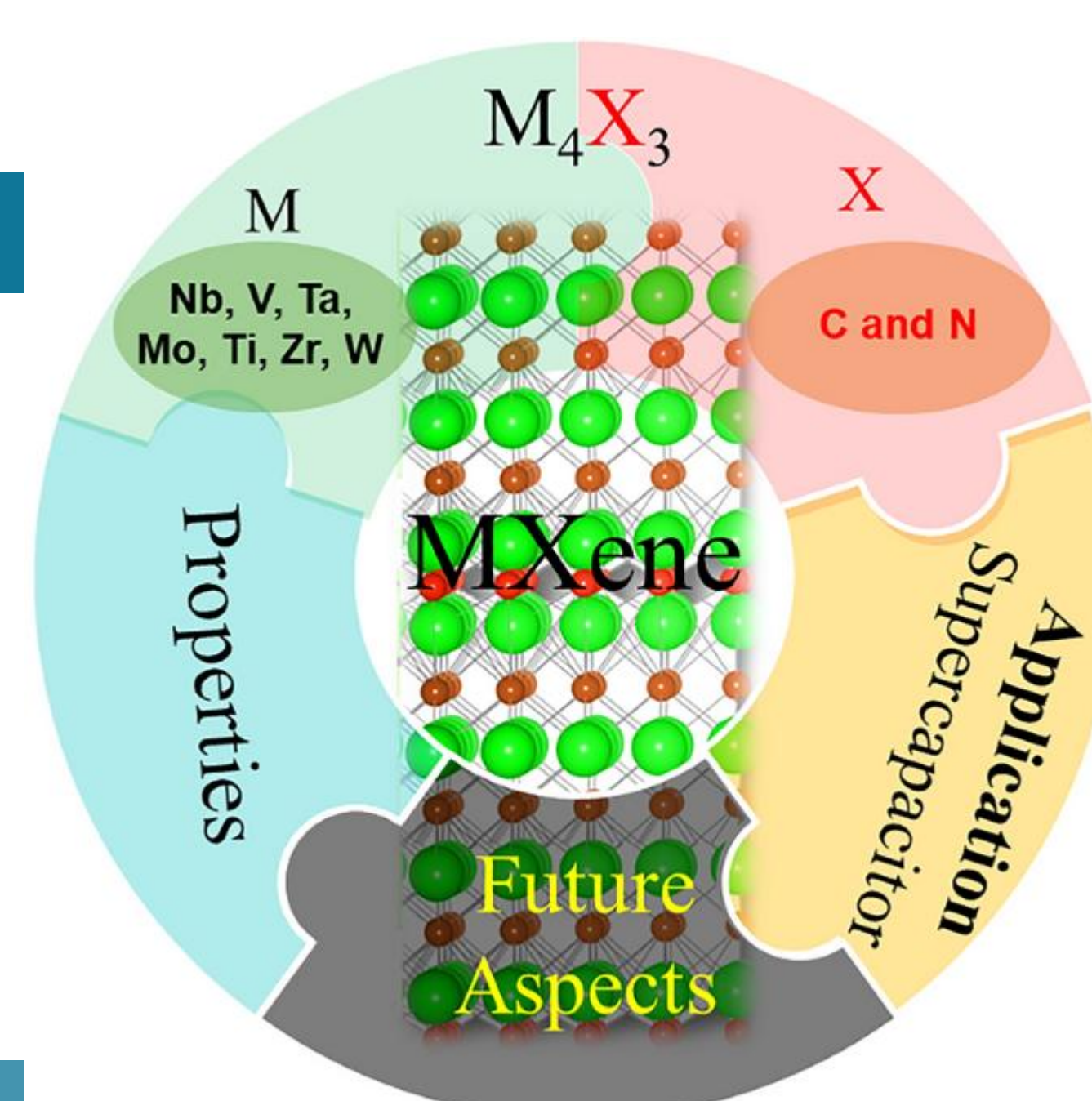
- Large surface area
- Tunable surface chemistry
- High conductivity

Applications

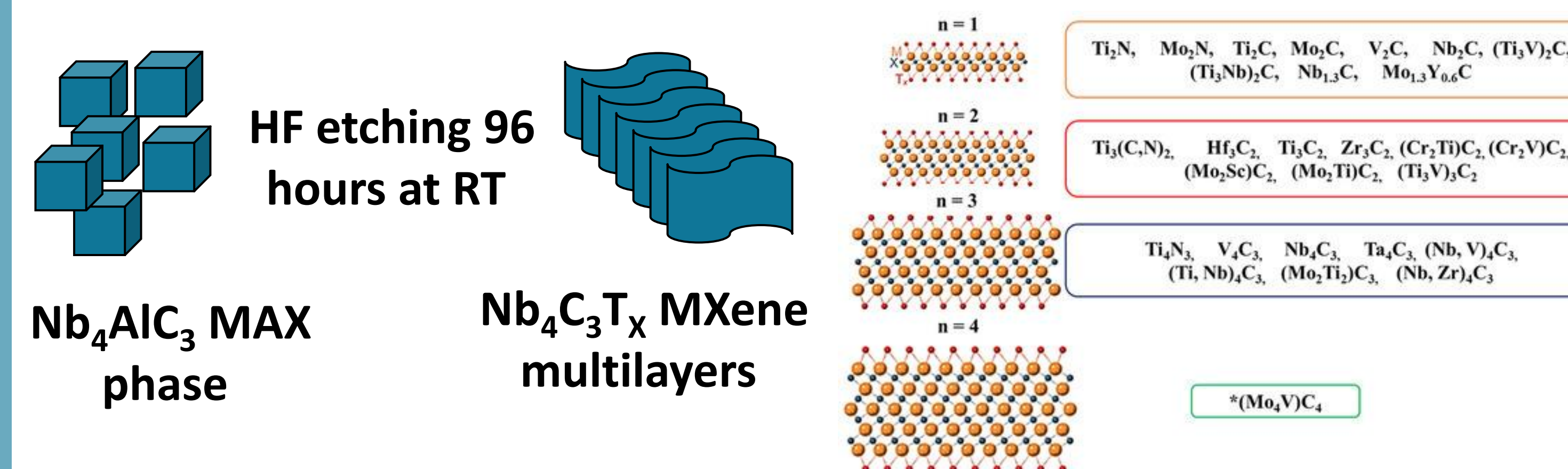
Sensors

EMI shielding

Electrocatalysis

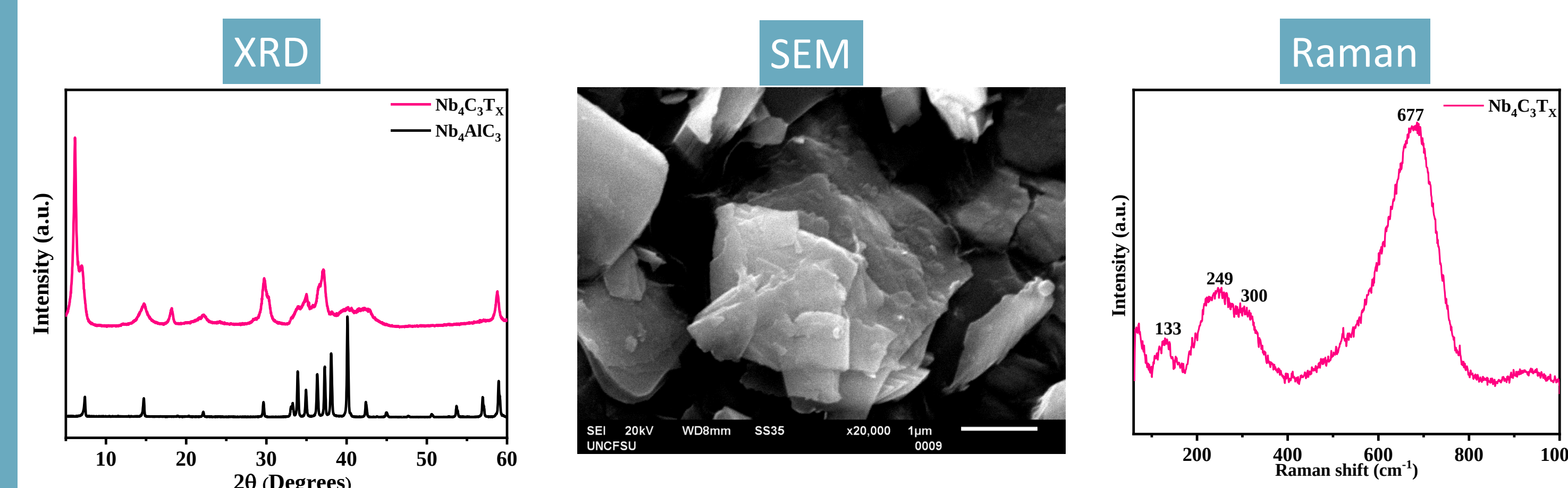


Synthesis and characterization of Nb₄C₃T_x MXenes



- 1 gram of Nb₄AlC₃ powder was treated with 10 mL of HF at RT for 96 hours with uniform stirring.
- After the reaction was complete, the resulting mixture was washed sequentially with hydrochloric acid (HCl) and deionized (DI) water until the pH of the solution reached approximately 6.
- The sample was then dried in an oven at 60 °C for 24 hours. The obtained MXene was subsequently used for various characterizations, including XRD, SEM, Raman spectroscopy, TGA, and DSC."

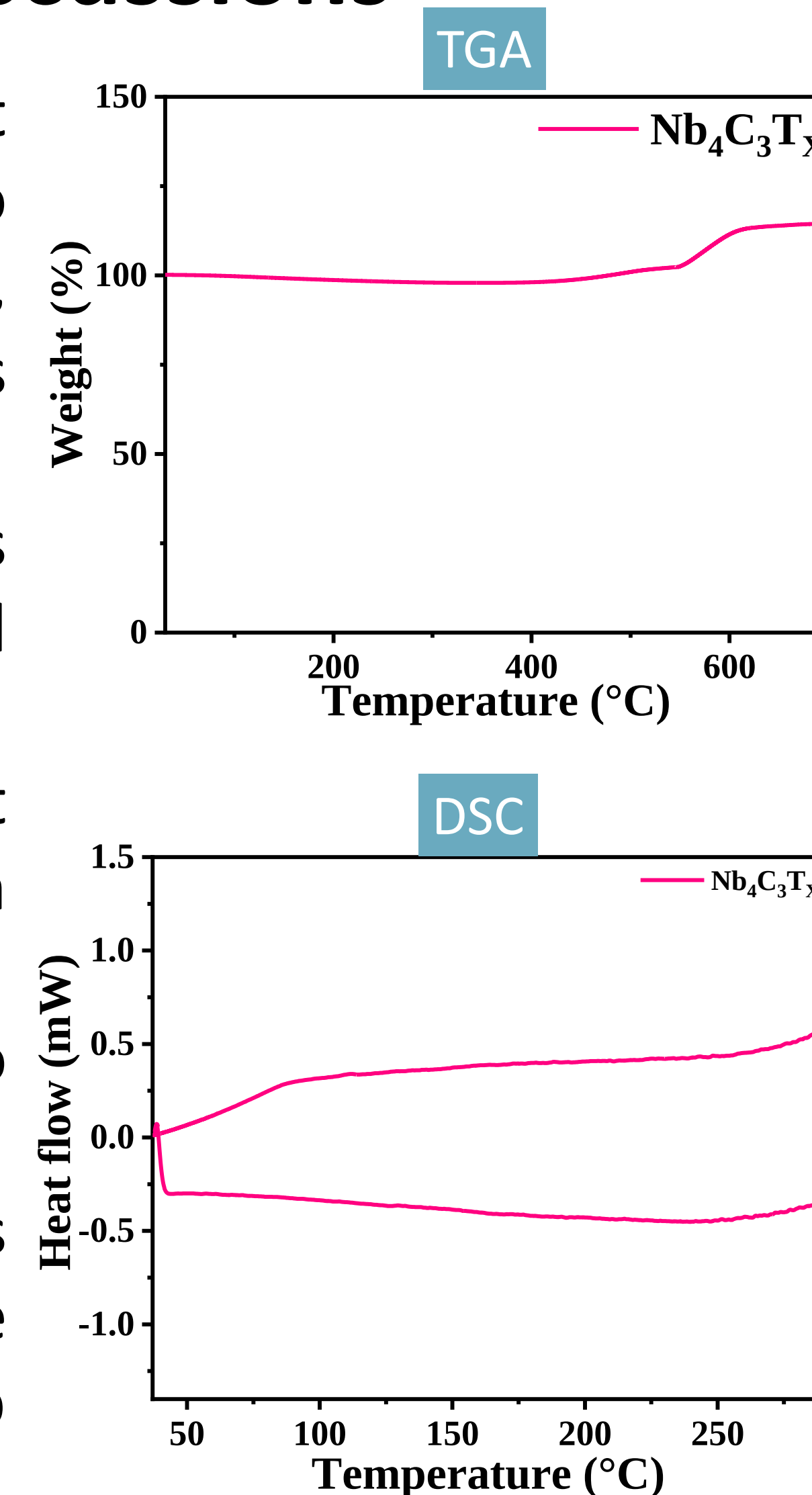
Results and Discussions



- The etching of Nb_4AlC_3 is confirmed by the appearance of a characteristic peak at 6.17° and 7.17° , corresponding to the (002) and (004) plane. A noticeable shift toward lower angles suggests an increased interlayer spacing due to sheet separation.
- SEM analysis further confirms the formation of $\text{Nb}_4\text{C}_3\text{T}_x$, revealing micro- and nanoscale sheet-like structures.
- The Raman peaks at show significant peaks at 133 cm^{-1} , Out-of-plane vibrations of niobium (Nb) atoms, 249 cm^{-1} : Nb-C stretching vibrations, 300 cm^{-1} : Mixed vibrations involving Nb-C bonds and Nb-T (T = surface terminations such as -O, -OH), 677 cm^{-1} : Vibrations associated with surface functional groups, particularly Nb-O or Nb-OH terminal bonds; may also indicate the presence of Nb_2O_5 due to surface oxidation.

Results and Discussions

- TGA, (Room Temp to $\sim 300^{\circ}\text{C}$): Slight weight loss ($\sim 1\text{--}2\%$) observed, due to Evaporation of adsorbed water, Possible desorption of surface groups (e.g., $-\text{OH}$, $-\text{F}$).
- ($300\text{--}500^{\circ}\text{C}$): The weight remains almost constant. Indicates high thermal stability.
- ($500\text{--}650^{\circ}\text{C}$): an increase in weight ($\sim 4\text{--}5\%$) is seen, Formation of niobium oxides (e.g., Nb_2O_5)
- DSC, Heat flow stabilizes, indicating no major phase transitions.
- Slight increase in heat flow, which is due to the Minor surface rearrangements of the functional group



Summary

- XRD and SEM analyses confirm the etching of $\text{Nb}_4\text{C}_3\text{Tx}$, as evidenced by the appearance of the characteristic etching peak and the sheet-like morphology.
- However, the appearance of an additional peak attributed to oxide formation suggests that Nb-based MXenes are susceptible to oxidation during synthesis, highlighting the need for further optimization of the etching process
- Raman spectroscopy also reveals signs of oxidation in the MXene structure.
- $\text{Nb}_4\text{C}_3\text{Tx}$ MXene shows good thermal stability up to **~500 °C**.

Acknowledgement

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References

- Lee, K., Kwon, S. Y., Choi, J., Kim, J., Woo, T., Ryu, J., ... & Lee, J. H. (2024). *Journal of Materials Chemistry C*, 12(3), 893-902.
- Lipatov, A., Alhabebe, M., Lu, H., Zhao, S., Loes, M. J., Vorobeva, N. S., ... & Sinitskii, A. (2020). *Advanced Electronic Materials*, 6(4), 1901382.
- Lan, L., Ni, Z., Zhao, C., Gao, J., Tang, X., Qu, Z., ... & Qiu, T. (2024). *Langmuir*, 40(40), 20945-20953.

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